Team Control Number

15276

Problem Chosen

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2024

HiMCM

Summary Sheet

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The addition and continuation of certain sports in the Olympics have become a heated topic in recent years. Viewed by millions, these sports can either bring more people toward the Olympic games or turn people away from joining in the fun. The Olympic Charter has already determined the basic criteria for sports to be included in the Olympics, but which are the best sports that should join or be reintroduced to the games? To help determine the sports that should be in the Olympics, our team has created a mathematical model that extensively accounts for variables affecting the likelihood of certain sports being added.

We begin by analyzing the factors/variables that impact the probability of sports, disciplines, or events (SDEs) being included/excluded based on the International Olympic Committee (IOC) criterias. We consider 78 different sports and consider 15 factor variables under IOC including number of players, Fatality Rate, Number of countries playing it, Ratio of Players of each gender, Cost of Equipment for Players, Social Media Presence, Ratio of each gender in coaching positions, Doping Statistics, and so on. These are often the strongest factor to influence the IOC to make the decision to select a sport for the Olympic Games.

For the overall model, we adopt an Analytic Hierarchy Process (AHP) to evaluate the relative importance of each variable for each individual sport. This subjective evaluation provides a process to start the decision making process. We also introduce the Entropy Method to analyze sports by using a decision matrix based on objective data to address the subjectivity. Ensembling the weights we calculate from both models, we then obtain the overall weighting used for each individual sport. These weights are then applied on the sport options to get scores. We utilize K-Means clustering to group sport into different groups to separate the sport. This unsupervised learning method provides us with three clusters. Finally, for the optimized model,we build a logistic regression machine learning (ML) model to incorporate the final score, cluster information and decision matrix factors for the probability including a sport. Then, the best probability of inclusion in each sport is output to our user. Consequently, we optimize our model by adjusting the values stored in the original decision matrix. While our model is not perfect, we believe it is a comprehensive system that takes sufficient factors into account. Furthermore, we have combined the AHP model, Entropy system, and our logistic regression model to develop an overall fairly realistic, effective, and objective mathematical model.

Keywords: Summer jobs; Analytic Hierarchy Process method; Entropy Weight method; K-means Clustering; Logistic Regression model

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1. Introduction

1.1 Background

What determines whether a sport should be in the Olympics or not? Truthfully, as the Olympic Charter only determines the basic requirements for sports to enter the Olympics, sports that should and should not be in the Olympics are intensely discussed, which involves balancing diverse and often conflicting interests such as cultural and historical significance, global representation, appealing to modern audiences, athlete safety and ethical concerns. In essence, the debate reflects the broader challenge of making the Olympics inclusive, modern, and sustainable while preserving its heritage and universal appeal. Every decision impacts athletes, fans, federations, and the identity of the Games, making it a complex and often polarizing issue.

To end this discussion, our team has created various mathematical models that provide in-depth analysis and outlines key factors to help decision-making. This paper analyzes relevant models such as AHP, Entropy method and Logistic regression in order to develop a sport evaluation system that recommends the most suitable sport to the IOC.

1.2 Problem Restatement

The primary goal is to create a model to evaluate sports for inclusion in the Olympics. This can be divided into these subquestions:

Question 1: Identify the factors the IOC considers for including or excluding a sport and gather data on these factors for common sports.

Question 2: Develop a recommendation system using the factors identified in Question 1. Assign weights to each factor for individual sports, ensuring the model is applicable to any sport.

Question 3: Apply the Entropy Method to reduce subjectivity in the recommendations from Question 2. Create an objective ranking of sports by combining subjective and objective results. Further refine the model using a Logistic Regression approach to integrate all considerations and derive final coefficients.

Question 4: Test the model with three sports recently added or removed from the Olympics; Three long-standing sports in the Olympics since 1988 or earlier; Identify and prioritize three potential sports for inclusion or reintroduction in the 2032 Brisbane Olympics; Provide recommendations for future Olympic Games in 2036 or beyond.

2. Assumptions and Variables

2.1 Assumptions and Justifications

Assumption 1: Accuracy and Reliability of Data and Inputs. It is assumed that all required data, including information on popularity, costs, and accessibility, sourced from the internet, is accurate and reliable. This assumption underpins the integrity of the evaluation process and the validity of the model's outcomes.

Assumption 2: Evaluation Based Exclusively on IOC Criteria. All sports are evaluated solely based on the established criteria of the International Olympic Committee (IOC), without the influence of external factors such as economic or political considerations, the balancing of tradition, host country preferences, or unforeseen events such as pandemics, wars, or safety concerns. All sports already follow the basic requirements of being in the Olympics games.

Assumption 3: Reasonability of Future Data Inputs. Future data inputs for the decision matrix, including trends derived from platforms such as Google Trends, are assumed to be accurate and reflective of public interest. This assumption facilitates more precise predictions and informed decision-making processes..

2.2 Variable Chart

2.2.1 Symbols and Definitions:

The table below defines all the variables we will use throughout this paper:

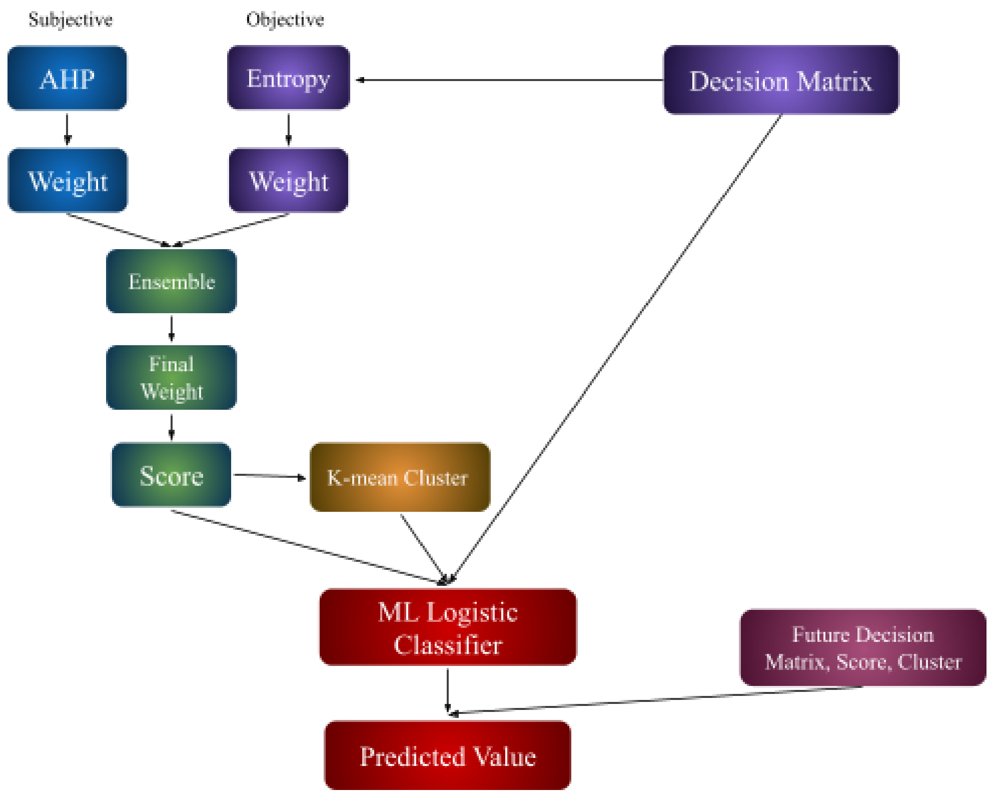
Variables Definition

Symbol	Definition
C	Comparison matrix
CI	Consistency index
RI	Random index
CR	Consistency ratio
Xij	Judgment matrix of job iii and evaluation of index jjj
Nij	Judgment matrix of job iii and evaluation after normalization
Pij	Weight of the iii-th plan of jjj-th index
ej	Entropy of jjj-th index
gj	Redundancy
W	Weight of the index
B	Regression coefficient for the model combining AHP and Entropy
k	K-mean cluster number

3 Model Description

3.1 General Process

Our valuation of the SDEs in the Olympics uses subjective and objective methodologies combined in an overall decision-making framework. First, the AHP assigned weight to the variables using subjective pairwise comparisons. These allow us to assess a set of factors, such as Fatality Rate, Gender Pay Gap, and CO2 Footprint, among others, in a proper structured hierarchical way. AHP is a method to help in simplifying complex decision factors, and it provides a rather intuitive framework for assessing the relative importance of each variable.



AHP is inherently subjective, and as such, it has been complemented by the Entropy Method, which is data-driven and computes weights according to the magnitude of fluctuation and distribution of quantitative data. This ensures that objective insights are added to the model in order to complement subjective evaluations. The generated weights from AHP and Entropy will be combined into a single final weight vector through the ensemble method by enhancing our evaluation in terms of reliability and robustness. With these final weights, a score is calculated for each SDE. The score will denote the suitability of the sport for the subjective and objective criteria. These scores are then processed through a K-means clustering algorithm, which clusters the SDEs together by their shared characteristics and performance on the evaluated attributes. Clustering enables the sports to fall into distinct groups, therefore bringing clarity in understanding their relative alignment to the IOC's values and objectives.

In parallel, we construct a decision matrix from the findings of the Entropy Method and the clustering outcomes. This matrix provides a compact summary of the whole evaluation process: in this format, it is relatively easy to visualize both subjective and objective data in one format. The decision matrix, along with scores and clusters, finally forms the input into the final stage of our analysis.

We apply a machine learning logistic classifier to make predictions of the inclusion or exclusion of SDEs. This model uses the scores, clusters, and decision matrix as independent variables to predict the likelihood of a sport being included in future Olympic Games. The logistic classifier provides a data-driven, probabilistic output that enhances decision-making transparency and accuracy. It will output the predicted values for each SDE, further refined decision matrices, scores, and clusters. In this way, the ensemble-based system ensures that our recommendations keep a strong connection with subjective expertise and objective data. Moreover, the methodology is an iterative process, and repeated application can be done in the future during evaluations as the landscape of Olympic sports changes over time. Through this approach, we aim to provide a robust, fair, and transparent process for guiding decisions about Olympic SDE inclusion.

3.2 Analytic Hierarchy Process (AHP) Model

3.2.1 General Process

The Olympic sports, disciplines, or events assessment starts with the identification of relevant variables and their hierarchical organization into a tree-like structure. This hierarchy ensures that the assessment of each SDE is systematic and clear.

To calculate the relative importance of these variables, we will use the AHP model, which arranges the variables in an intuitive hierarchy for pairwise comparisons. AHP enables the simplification of complex decision-making by structuring the variables and weighting according to subjective evaluations without complicating effective decision-making. The summary of the hierarchy structure of our variables is given as follows:.

In this respect, we adopt the entropy method as a complementing method to AHP to quantify subjective judgment. The Entropy Method objectively assigns weights based on the variation and dispersion of quantitative data of each variable. This dual approach balances the subjectivity of AHP with the objectivity of the entropy-based data analysis, thus increasing the accuracy and reliability of our model.

The final model uses a machine learning approach to integrate the strengths of AHP and the Entropy Method. The process involves the following steps:

1. Calculating scores of sport using combined weights received from AHP and the Entropy Method.
2. Grouping the SDEs into clusters of their scores to result in similar attributes.
3. Formulating decision matrices based on insights derived from entropy, which can be used as independent variables under a logistic regression framework.

The dependent variable in this model is the inclusion or exclusion decision for each SDE. Independent variables are created from the combined AHP-Entropy scores and further IOC-aligned quantitative data. This ensembles-based approach optimizes the recommendation for accuracy and fairness; thus, the decisions are contextually relevant and data-driven.

While AHP allows for a structured evaluation, its subjective nature necessitates mitigation. By combining AHP with the Entropy Method, we leverage both expert judgment and objective data to create a robust evaluation framework. For instance, Entropy-derived weights account for the variability in quantitative factors, reducing biases in AHP's pairwise comparisons. It ensures that decisions concerning the inclusion or exclusion of Olympic SDEs are made with transparency and equal weight given to both subjective considerations and objective data. The flowchart below describes the logical sequence of the steps in the process, with a detailed hierarchy structure of our variables summarized for reference.

Number Index	**Definition**	**IOC Criteria It Belongs To**
1	**Fatality Rate** We believe the Fatality Rate of a sport helps determine how dangerous a sport is. If a sport is more dangerous, its safety will go down, going against the IOC's criteria. Therefore, a sport with a higher fatality rate will rank lower on our model.	Safety
	Measured in number of deaths per million	
2	**Injury Rate** Injury rate is similar to Fatality Rate as results in the safety of a sport going down. As Safety is one of our priorities, we must consider the injuries that may occur to athletes when performing a certain sport. However, as fatalities are worse than injuries, the injury rate is weighed less in our model.	Safety
	Measured in average number of injuries per 1000 players	
3	**Number of Players** The number of players heavily informs us about the popularity of a sport. As the Olympics want to be inclusive and accessible, more popular sports with more people playing it will be more suitable for the Olympic Games.	Popularity
	Measured in millions	
4	**Social Media Presence** Social Media is very popular in the current day and age. The presence of a certain sport's presence on social media can, therefore, help determine its popularity. Moreover, as social media is primarily used by youth, social media prescience can also help us better understand the sports relevant to today's younger audience.	Popularity
	Measured on a scale of 1-10 in relation to each other	
5	**Movies/Documentaries** Movies and documentaries show a sport's cultural importance and emotional appeal. They raise global awareness and highlight Olympic values like teamwork and perseverance. Popular films engage younger audiences and inspire people, aligning with the Olympic spirit, which makes the sport more relevant to general people.	Relevance
	Measured using the number of movies or documentaries that are easily found online.	

6	**Accessibility across different streaming platforms** Streaming is widely used, especially by younger viewers, making it a key tool for promoting sports and ensuring inclusivity. It aligns with the IOC's goal of uniting people worldwide through sports. Additionally, viewership data from these platforms helps prove a sport's popularity and relevance, keeping the Olympics modern and accessible. Measured using the number of accessible streaming platforms	Accessibility
7	**Number of Countries Playing** The number of countries playing a certain sport is crucial in determining popularity as well as accessibility. If more countries are able to play the sport, that means it is easily set up and played by anyone. Also, the Olympic rings symbolize the unity of countries in sports, meaning sports played in more countries align better with Olympic values. This factor also heavily aligns with the IOC Criteria of Inclusivity where 75 countries must play the sport in order for it to be eligible to join the Olympics. Measured by the number of countries playing the sport from Olympic Statistics	Popularity
8	**Presence in University Sports** This factor shows its potential for growth and innovation, aligning with the IOC’s focus on modern and forward-thinking sports. University programs drive new training methods, engage younger generations, and create a pipeline of future athletes. They also promote diversity and broad participation, supporting Olympic values while ensuring the sport’s global appeal. Measured by how many university sport teams play it worldwide	Innovation
9	**Cost of Equipment** The cost of equipment can hinder individuals from playing certain sports, resulting in that sport having lower accessibility. Whereas, if pricing for equipment is cheaper, more people will be able to purchase equipment and play the sport, leading to higher popularity and inclusivity. We assume all costs in different countries are relatively the same. Measured using the average price per person for equipment ($)	Inclusivity
10	Ratio of Players by Gender The Olympic Games heavily value gender equity across their sports. The ratio of players by gender aids in illustrating whether women are able to join into the sport relatively easily, with a more equal ratio resulting in higher Male to Female ratio	Gender Equity

11	**CO2 Footprint** CO2 Footprints help determine the environmental harms certain sports could have on the environment. Therefore, we measure the footprint of each sport, comparing them with each other and creating a ranking system. The higher the carbon footprint, the more harmful the sport is to the environment. Whereas a lower carbon footprint results in a less environmentally harmful sport. Measured on a ranking system with 1 being the lowest emissions and 100 being the highest	Sustainability
12	**Gender Pay Gap** The gender pay gap shows how fair a sport is to male and female athletes. A smaller gap means better equality and opportunities for everyone. Rated 1 through 10 where 1 is the least and 10 is the most	Gender Equity
13	**Gender Ratio in Coaching** The gender ratio in coaching shows how well a sport supports equal opportunities in leadership. A balanced ratio promotes fairness, diversity, and breaks stereotypes. Sports with fair coaching representation align with the IOC's goal of gender equity.The ratio close to 1 indicates balanced gender participation and good gender equity. If the ratio is close to 5, it indicates the sport is one gender dominated one such as baseball. Defined as a ratio between 1 to 5 where 1 is the least and 5 is most	Gender Equity
14	**Doping Statistics** Doping statistics show how well a sport upholds fair play. Low doping incidents mean the sport enforces rules and promotes honest competition, building trust among athletes and audiences. Sports with good doping records support the Olympic values of fairness and integrity. Measured in percent found doping based on Olympic Statistics	Fair Play
15	**Ease of Developing New Strategies** Sports that allow for new strategies show adaptability and creativity, aligning with the IOC's focus on innovation. They keep the sport dynamic, engaging, and exciting for players and audiences, especially younger ones. This flexibility ensures the sport stays relevant and competitive in the Olympic program. Rated 1 through 5 to show ease of developing new strategies with 1 being the least and 5 being the most.	Innovation

3.2.2 Weight Vector

We use the Analytic Hierarchy Process to obtain the weight vector. AHP is especially useful when it is difficult to create a weight vector directly from data provided by research. By performing pairwise comparisons and offering the ability to determine how consistent these comparisons are, AHP provides a result that is generally rational and trustworthy.

In order to make these comparisons, we use Saaty's fundamental scale of absolute numbers. The scale assigns values as in the following:

1-9 Scale Definitions

Scale	Definition
1	When two attributes are equally important.
3	When one attribute is moderately more important than the other.
5	When one attribute is strongly more important than the other.
7	When one attribute is very strongly more important than the other.
9	When one attribute is extremely more important than the other.

The intermediate values, 2, 4, 6, 8 represent situations that fall between those that result in these judgments above.

Based on this 1-9 scale, it is possible for us to establish a comparison matrix for the variables. A comparison matrix is a square matrix of the following form, as presented in the following tables. This would form the base for calculating the weight vector.

Technology Based SDE's:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	2	5	4	4	3	7	2	3	9	7	7	7	3	3
2	1/2	1	1/3	1/3	1/3	1/2	1/4	1/5	1/3	1/2	1/2	2	2	1/3	1/4
3	1/5	3	1	1/3	1	2	2	1/3	1	4	4	6	6	1/2	1/3
4	1/4	3	3	1	1/4	1/2	1/3	1/6	1/4	1	1	2	2	1/5	1/4
5	1/4	3	1	4	1	2	5	1/2	4	5	5	6	5	2	2

6	1/3	2	1/2	2	1/2	1	1/2	1/4	1/3	3	3	3	2	1/4	1/4
7	1/7	4	1/2	3	1/5	2	1	1/2	1/3	4	4	5	3	1/3	1/2
8	1/2	5	3	6	2	4	2	1	3	6	6	7	5	2	2
9	1/3	3	1	4	4	3	3	1/3	1	5	5	6	4	2	2
10	1/9	2	1/4	1	1/5	1/3	1/4	1/6	1/5	1	1	2	1	1/4	1/4
11	1/7	2	1/4	1	1/5	1/3	1/4	1/6	1/5	1	1	1	1	1/3	1/3
12	1/7	1/2	1/6	1/2	1/6	1/3	1/5	1/7	1/6	1/2	1	1	1/2	1/5	1/5
13	1/7	1/2	1/6	1/2	1/5	1/2	1/3	1/5	1/4	1	1	2	1	1/5	1/5
14	1/3	3	2	5	1/2	4	3	1/2	1/2	4	3	5	5	1	1/2
15	1/3	4	3	4	1/2	4	2	1/2	1/2	4	3	5	5	2	1

Team Based SDE's:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	9	3	4	3	2	5	2	3	4	4	3	4	4	4
2	1/9	1	1/5	1/3	1/5	1/2	1/2	1/5	1/2	1/3	1/3	1/4	1/4	1/4	1/4
3	1/3	5	1	3	1	2	1	1/2	1/3	3	3	4	4	4	4
4	1/4	3	1/3	1	1/2	2	1/2	1/3	1/3	1	1	3	3	3	3
5	1/3	5	1	2	1	2	1	1/3	1/3	2	2	3	3	3	3
6	1/2	2	1/2	1/2	1/2	1	1/2	1/3	1/3	1/2	1/2	2	2	2	2
7	1/5	2	1	2	2	2	1	1/2	1/2	2	2	4	4	4	4
8	1/2	5	2	3	3	3	2	1	1	2	2	4	4	4	4
9	1/3	2	3	3	3	3	2	1	1	2	2	4	4	4	4
10	1/4	3	1/3	1	1/2	2	1/2	1/2	1/2	1	1	3	3	3	3
11	1/4	3	1/3	1	1/2	2	1/2	1/2	1/2	1	1	3	3	3	3
12	1/4	4	1/4	1/3	1/3	1/2	1/2	1/4	1/4	1/3	1/3	1	1	1	1
13	1/4	4	1/4	1/3	1/3	1/2	1/2	1/4	1/4	1/3	1/3	1/3	1	1	1
14	1/4	4	1/4	1/3	1/3	1/2	1/2	1/4	1/4	1/3	1/3	1/3	1	1	1

15	1/4	4	1/4	1/3	1/3	1/2	1/2	1/4	1/4	1/3	1/3	1/3	1	1	1

Individual SDE's:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	4	4	4	4	4	4	4	4	4	4	4	4	4	4
2	1/4	1	1/2	1	1/2	2	1/2	1/3	1/3	1	1	2	2	2	2
3	1/4	2	1	3	1	2	1	1/2	1	3	3	4	4	4	4
4	1/4	1	1/3	1	1/2	1	1/2	1/3	1/3	1	1	2	2	2	2
5	1/4	2	1	2	1	3	1	1/2	1/2	2	2	4	4	4	4
6	1/4	1/2	1/2	1	1/3	1	1/2	1/4	1/3	1	1	2	2	2	2
7	1/4	2	1	2	1	2	1	1/2	1/2	2	2	3	3	3	3
8	1/4	3	2	3	2	4	2	1	2	3	3	4	4	4	4
9	1/4	3	1	3	2	3	2	1/2	1	2	2	3	3	3	3
10	1/4	1	1/3	1	1/2	1	1/2	1/3	1/2	1	1	2	2	2	2
11	1/4	1	1/3	1	1/2	1	1/2	1/3	1/2	1	1	2	2	2	2
12	1/4	1/2	1/4	1/2	1/4	1/2	1/3	1/4	1/3	1/2	1/2	1	1/2	1/2	1
13	1/4	1/2	1/4	1/2	1/4	1/2	1/3	1/4	1/3	1/2	1/2	2	1	1	2
14	1/4	1/2	1/4	1/2	1/4	1/2	1/3	1/4	1/3	1/2	1/2	2	1	1	2
15	1/4	1/2	1/4	1/2	1/4	1/2	1/3	1/4	1/3	1/2	1/2	1	1/2	1/2	1

1 versus 1 Fighting SDE's:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	4	3	3	3	4	3	3	3	3	3	4	4	4	4
2	1/4	1	1/2	1/2	1/2	1	1/2	1/4	1/3	1/2	1/2	1	1	1	1
3	1/3	2	1	2	1	3	1	1/4	1/2	2	2	3	3	3	3
4	1/3	2	1/2	1	1/2	2	1/2	1/4	1/3	1	1	3	3	3	3
5	1/3	2	1	2	1	2	1	1/3	1/2	2	2	3	3	3	3

6	1/4	1	1/3	1/2	1/2	1	1/2	1/4	1/3	1/2	1/2	2	2	2	2
7	1/3	2	1	2	1	2	1	1/3	1/2	2	2	3	3	3	3
8	1/3	4	4	4	3	4	3	1	2	3	3	4	4	4	4
9	1/3	3	2	3	2	3	2	1/2	1	2	2	3	3	3	3
10	1/3	2	1/2	1	1/2	2	1/2	1/3	1/2	1	1	3	3	3	3
11	1/3	2	1/2	1	1/2	2	1/2	1/3	1/2	1	1	3	3	3	3
12	1/4	1	1/3	1/3	1/3	1/2	1/3	1/4	1/3	1/3	1/3	1	1/3	1/2	1/2
13	1/4	1	1/3	1/3	1/3	1/2	1/3	1/4	1/3	1/3	1/3	3	1	2	3
14	1/4	1	1/3	1/3	1/3	1/2	1/3	1/4	1/3	1/3	1/3	2	1/2	1	1
15	1/4	1	1/3	1/3	1/3	1/2	1/3	1/4	1/3	1/3	1/3	2	1/3	1	1

1 versus 1 Non-fighting SDE's:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	4	3	3	3	4	3	2	3	3	3	4	4	4	4
2	1/4	1	1/2	1/2	1/2	1/3	1/2	1/3	1/3	1/2	1/2	1/3	1/3	1/3	1/3
3	1/3	2	1	1	1/3	1/2	1/3	1/2	1/2	1	1	1/2	3	3	3
4	1/3	2	1	1	1/2	1	1/2	1/3	1/3	1	1	1/2	1/2	1/2	1/2
5	1/3	2	3	2	1	1	1	1/2	1/2	1	1	1/2	1/2	1/2	1/2
6	1/4	3	2	1	1	1	1/2	1/3	1/3	1/2	1/2	2	1	1	1
7	1/3	2	3	2	1	2	1	1/2	1/2	1	1	1/2	1/2	1/2	1/2
8	1/2	3	2	3	2	3	2	1	2	1/2	1/2	1/3	4	4	4
9	1/3	3	2	3	2	3	2	1/2	1	1	1	1/2	3	3	3
10	1/3	2	1	1	1	2	1	2	1	1	1	1/2	1/2	1/2	1/2
11	1/3	2	1	1	1	2	1	2	1	1	1	1/2	1/2	1/2	1/2
12	1/4	3	2	2	2	1/2	2	3	2	2	2	1	1/3	1/2	1/2
13	1/4	3	1/3	2	2	1	2	1/4	1/3	2	2	3	1	1	1/2

14	1/4	3	1/3	2	2	1	2	1/4	1/3	2	2	2	1	1	1
15	1/4	3	1/3	2	2	1	2	1/4	1/3	2	2	2	2	1	1

Each SDE has a different weight for each factor. For example, in boxing, the injury rate might be weighted higher than in other sports, as injury is more of a concern in boxing. Since we had 15 factors for each SDE with each factor having some relation to the IOC criteria, that would correspond to a 15 by 15 matrix for each SDE. Each of the weights were on a scale from 0 to 9, with weights from 1 to 9 indicating the row factor is more important than the column factor, and a weight from 1/9 to 1 indicating the column factor is more important. Since it was not feasible to create a different matrix for each SDE, we categorized all SDE's into 5 main categories, and used category specific comparison matrices.

The second step is to create a decision matrix for each SDE. This would mean actually researching to find the values for each factor. For example, we would find the fatality rate for each SDE. We then normalized the decision matrix because there were many different factors with different units. The normalization formula depended on whether the "desirable" outcome was a high statistic (e.g. number of people playing the SDE) or a low statistic (e.g. fatality rate).

For factors where a high statistic was desirable, we used

$$N_{ij} = 0.998\frac{X_{ij} - min\{X_{1j}, \dots, X_{nj}\}}{max\{X_{1j}, \dots, X_{nj}\} - min\{X_{1j}, \dots, X_{nj}\}} + 0.002$$

On the other hand, if a low statistic was desirable, we used

$$N_{ij} = 0.998\frac{max\{X_{1j}, \dots, X_{nj}\} - X_{ij}}{max\{X_{1j}, \dots, X_{nj}\} - min\{X_{1j}, \dots, X_{nj}\}} + 0.002$$

Once these two matrices are completed, we can calculate the suitability index for each SDE by multiplying together the two matrices.

Using this method, we integrate subjective expert opinions to rank the weight of each variable. By applying the AHP model, we calculate a suitability index for each SDE, allowing us to rank and compare them. This approach ensures that all SDEs are evaluated against a consistent baseline, guiding recommendations for the 2032 Olympic Games.

3.3 Using the Entropy Weight Method

To decrease the subjectivity of the weighting present in our original AHP model, we also employed the Entropy Weight Method (EWM). The EWM provides a more objective weighting as the weights are based on the variability or "entropy" of the data, which is objective. Factors that have a higher variability receive higher weights, while factors with a lower variability receive lower weights.

To ensure the consistency of the input data and validate the reliability of the model, we assess the consistency of an $n \times n$ comparison matrix using its largest eigenvalue (λ_{max}). The Consistency Index (CI) is calculated using the formula:

$$CI = \frac{\lambda_{max} - n}{n - 1}$$

After calculating the CI, we introduce the Random Consistency Index (RI) and the Consistency Ratio (CR) to evaluate whether the level of inconsistency is acceptable. The CR is calculated as:

$CR = \frac{CI}{RI}$

The threshold for acceptable inconsistency is $CR < 0.1$. Using these formulas, we evaluate our comparison matrix which are all under 0.1.

3.4 Integrating the AHP Model and Entropy Weight Method Together

To reduce subjectivity and improve the accuracy and reliability of our model, we integrate the subjective AHP approach with the objective Entropy method. This integration addresses the inherent subjectivity of the AHP model by balancing it with the data-driven nature of the Entropy method. To achieve this, we utilize the following formula, designed to minimize the deviation between the AHP and Entropy systems:

$$B = min \sum_{j=1}^{n} (\alpha W_A - \beta W_E)^2$$

$$s.t\ \alpha + \beta = 1$$

where α and β are constants representing the proportional influence of AHP and Entropy. W_A represents the weight derived from the AHP model and W_E represents the weights derived from the EWM. By applying ordinary least squares, we calculate the regression coefficient B, which indicates the relationship between W_A and W_E. Using this, we determine the proportions $u1$ and $u2$ to combine W_A and W_E:

$$u1 = \frac{1}{1+B},\ u2 = \frac{B}{1+B}$$

$$W_f = u1 \cdot W_A + u2 \cdot W_E$$

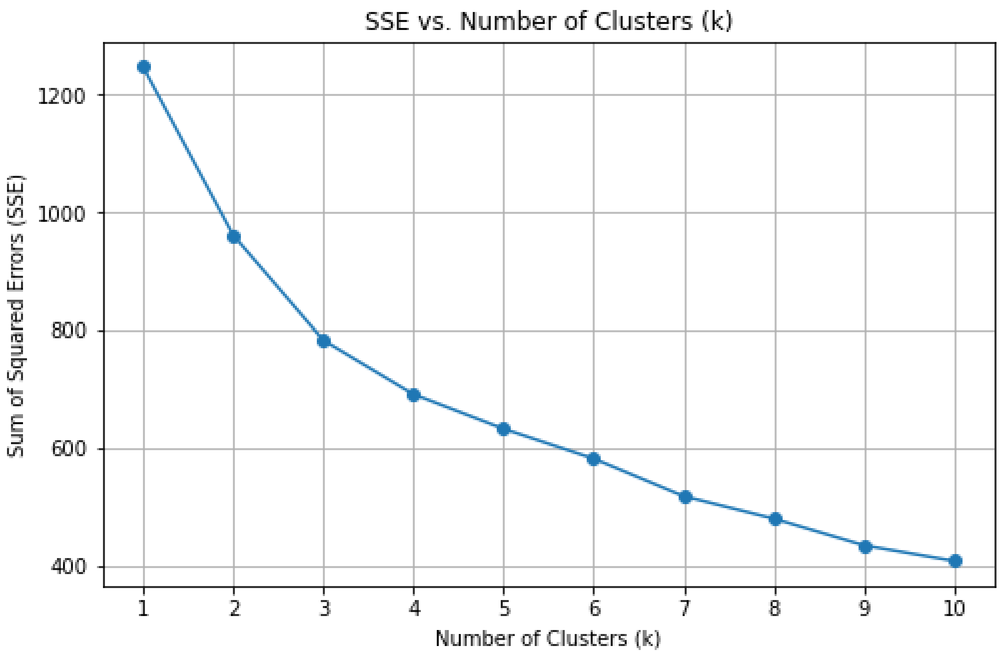
Here, W_f represents the final optimized weight for each criterion. This combined weight reflects both subjective preferences and objective evaluations, ensuring a balanced and robust result for the index.

3.5 K-means Clustering

K-means clustering [9] is used to group sports into categories based on their characteristics. It is simple, scalable, fast, and works well with sparse data. The Euclidean Distance formula measures similarity between clusters, and the Sum of Squared Errors (SSE) helps determine the best number of clusters (K) for accurate grouping.

$$d(a, b) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

As the number of cluster k increases, the clusters are divided into more precise groups, leading to a higher level of clustering. Consequently, the SSE (Sum of Squared Errors) gradually decreases. When k is below the optimal number of clusters K, SSE decreases significantly because adding clusters improves the grouping and reduces error. However, once k reaches the optimal number of clusters, the improvement in clustering diminishes. The SSE decline slows down and approaches a constant value as k continues to increase. The graph below illustrates the relationship between SSE and k, based on the final sport score.



From the graph, it is clear that SSE decreases as the k value increases, with the most significant gradient change occurring between k=2 and k=3. Therefore, k=3 is selected as the optimal number of clusters. After applying K-means clustering, we obtain the recommended list of sports groups, corresponding to the clusters with the group for continuous staying Olympic Games and potential remove/added sport. The figure 2(Left) shows one of the clusters with sports consistently appearing in the Olympics. The Result Score is also displaced as an indicator of strong standing if the number is bigger. The figure 2(Right) shows the lowest Sport Results Scores and their associated clusters. There is still no clear cut for sports that should be removed from the Olympics. Therefore, we develop an optimal ensemble model using the Logistic Regression model to improve the accuracy.

K-Mean Clustering results and Sport Result Scores:

Label	Cluster	Result	ID
Baseball	1	0.61392	10
Basketball	1	0.71572	11
Cricket	1	0.62162	12
Field Hockey	1	0.6555	13
Football	1	0.73552	15
Volleyball	1	0.71987	20
Tennis	1	0.73076	43
Swimming	1	0.72274	61
Cycling	1	0.59921	63

Label	Cluster	Result	ID
Enduro Motorcycle R	2	0.25993	77
Motocross	2	0.3101	76
Boxing	2	0.35017	25
Water Motorsports	2	0.37725	78
Equestrian	0	0.38955	64
Hurling	0	0.40952	4
Drag Racing	2	0.41253	75
Sumo	3	0.43192	24
Kickboxing	3	0.43767	22
Gaelic Football	0	0.43852	3

3.6 Logistic Regression Model

K-mean cluster is unsupervised machine learning and it has less prediction power compared to supervised learning. We choose Logistic Regression as a supervised machine learning algorithm used for binary classification problems. It predicts the probability that a sport should be included in the Olympic Games or not (1=Yes, 0=No). Logistic regression is the process of modeling the probability of a discrete outcome given an input variable [7]

It takes the following form:

$$LR(x; b) = \frac{1}{1+e^{-b_0 - \sum_{i=1}^{n} b_i x_i}}$$

Where $b = [b_0, b_1, \dots, b_n] \in R^{n+1}$ is the vector for the constant term and coefficients of n input variables, $x_0, x_1, x_2, \dots, x_n$. Here we have n=17 input variables in this study listed as factor 1 to factor 15 in section 3.1 in decision matrix, Sport Score based on ensembled weights from AHP and Entropy, and Cluster result based on K-mean clustering. Cluster variable is a categorical variable and we have encoded it before running the logistic model. The learning process involves computing the appropriate values of b, which maximize the log likelihood. [7] Python program was used to estimate the coefficients of this model.

4 Testing the Models

4.1 Results and Analysis

Q2: Use your factors to build a model (or set of models) to help the IOC evaluate which SDEs align best with Olympic criteria. Based on the coefficient, the top three most important factors are the number of countries playing the sport, the ratio of each gender in coaching positions and social media presence. We have regression coefficients estimated as following:

Feature	Coefficient
3 Number of countries playing it	1.0769126
11 Ratio of each gender in coaching positions	0.8698298
Cluster_3	0.849821
9 Social Media Presence	0.6809792
Cluster_1	0.6146277
4 Ratio of Players of each gender	0.4424704
14 Accessibility across different streaming platforms	0.3608114
15 Movies/documentaries	0.3478957
7 Presence in University Sports	0.3232432
2 CO2 footprint	0.3087879
5 Number of Players	0.2845262
Score	0.283693
1 Fatality Rate	0.282829
6 Cost of Equipment for Players	-0.083562
8 Injury Rate	-0.091292
Cluster_2	-0.197364
10 Gender Pay Gap	-0.449553
12 Doping Statistics	-0.697894
13 Ease of developing new strategies	-0.788798
Cluster_0	-1.267122

Q3: Test your model on at least three SDEs that have been added or removed from recent Olympics, namely Olympic years 2020, 2024, and 2028 and at least three SDEs that have continuously been in the Olympic program since the 1988 games or earlier.

Label	True Label	Predicted Label	Probability Positive Class	Type
Breaking	0	1	0.549585	Newly added
Climbing (Bouldering, Traditional Climbing)	0	1	0.571432	Newly added
Surfing	0	1	0.552365	Newly added
Basketball	1	1	0.947977	Continuous
Swimming	1	1	0.923738	Continuous
Football	1	1	0.915329	Continuous
Boxing	1	0	0.478453	Removed

Our model can accurately predict the sports that have been added or removed from recent Olympics such as Breaking, Climbing, Surfing and Boxing. Mosel shows a very high probability for continuous sports such as basketball, swimming, football. Again, probability more than 0.5 would be defined as an Olympic sport. Boxing has been debatable because it has a probability of 0.47 which means it can be very likely to be added back to Olympic in the future.

Our model evaluates sport based on the criteria defined by the IOC, such as popularity, gender equity, sustainability, accessibility, and innovation. Example: A high score in global popularity or inclusivity metrics supports the inclusion of a sport. Consistency with environmental sustainability criteria, such as low CO2 emissions, reinforces its compatibility with Olympic goals. Our model indicates that swimming aligns with the IOC's values due to its sustained global participation and evidenced by strong media representation and consistent athlete participation from many countries over the years.

Our model used ranking, weights, scores, clustering grouping and regression to demonstrate how a sport fit into its optimal categories and quantify their performance relative to key IOC criteria. The probability for continuous sport is aligned with the high score sports. This is a data driven approach to support decision making.

Baseball has exceptionally high values in gender inequity, in this nearly male dominated sport, which receives assignment as a non-Olympic sport, affirming its alignment with the IOC's emphasis on gender equity.

While squash has not been an official Olympic Game, its high scores in growth of players, innovation and youth engagement justify its future Olympic status. Same for breaking, it has high scores in innovation and youth engagement which justify its future Olympic status.

Q4. Identify three SDEs that could be new additions or reintroductions for the 2032 Olympics in Brisbane. Make sure to identify which SDE should be considered first, second and third for inclusion in the Brisbane games. Are there any SDEs that you believe have potential for inclusion in an Olympic games in 2036 or beyond?
We adjusted the decision matrix to generate the predicted value for each factor, updated the scores and clusters for 2032 and 2036.

Label	True Label	Predicted	Probabilit
Baseball	0	1	0.75932
Beach Soccer	0	1	0.752939
Cricket	0	1	0.749991
Squash	0	1	0.745678
Kickboxing	0	1	0.586026
Mixed Martial Arts (MMA)	0	1	0.533281
Martial Arts (Karate, Kung Fu)	0	1	0.506904
Boxing	1	0	0.478453
Sailing	1	0	0.166014
Coastal Rowing	1	0	0.130612
Billiards (Pool)	0	0	0.496102

According to our model, Baseball may be added back to the Olympics. Beach Soccer, Cricket, Squash are likely to be added into the 2032 Olympics. In 2036, potential sports are Kickboxing, MMA and Martial Arts.

5 Evaluation of Model

We used 78 sports excluding baseball to develop a logistic regression model using a machine learning approach and we fit the model with balanced accuracy 0.82. This means our model has about an 18% chance to go wrong. There were 16 sports used as a test dataset. To determine the accuracy of our model we use the following equation:

$$Accuracy = \frac{TP+TN}{TP+FP+FN+TN}$$

Where TP, FP, FN, TN, come from the following confusion matrix.

		Prediction	
		Success	Failure
Truth	Success	TP (7)	FN (2)
	Failure	FP (1)	TN (6)

We also define the model evaluation metrics as following:

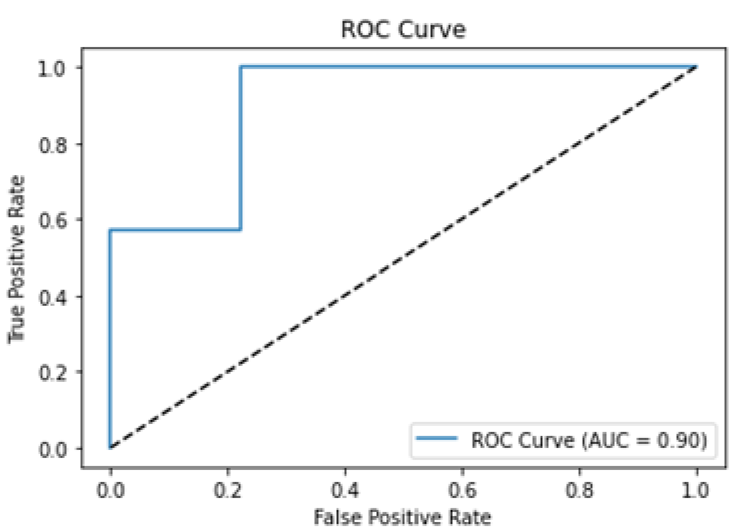
$$Sensitivity = \frac{TP}{TP+FN}$$

$$Specificity = \frac{TN}{FP+TN}$$

$$Balanced\ Accuracy = \frac{\frac{TP}{TP+FN}+\frac{TN}{FP+TN}}{2}$$

Sensitivity refers to our model's ability to identify a sport as an Olympic game. A highly sensitive test means that there are few false negative results, and thus fewer cases of good sport are missed. The specificity of our model is its ability to identify a sport which should be removed from the Olympic Games as negative. A low specificity means that there are more removable sports that are identified incorrectly. In the 16 sports, our model can correctly identify 7 sports in the Olympics. And 6 non-Olympic sports. Only 3 sports are incorrectly recommended. This gave us sensitivity=0.86 and specificity=0.78. This means that our model did better to pick up sport for the Olympic game in 2028.

Figure 6: ROC Curve and AUC



A Receiver Operator Characteristic (ROC) curve is a graphical plot used to show the model fit ability. One common approach is to calculate the area under the ROC curve, which is abbreviated to AUC. It is used to measure how well the model predicts and performs. AUC is close to 1 indicating a better model. We had AUC=0.90 based on our data indicating an excellent model fit.

5.1 Sensitivity Analysis

In our model, the impact factors are assessed as $100p$ times a linear combination of attribute values. The sensitivity of the model is analyzed as follows:

For the variable p, when other variables are held fixed, reducing p from 1.0 to 0.9 will result in a 90% decrease from its current impact factor value.
For a given attribute value, with other variables held constant, the change in the impact factor is proportional to the change in the attribute value. It is represented as:

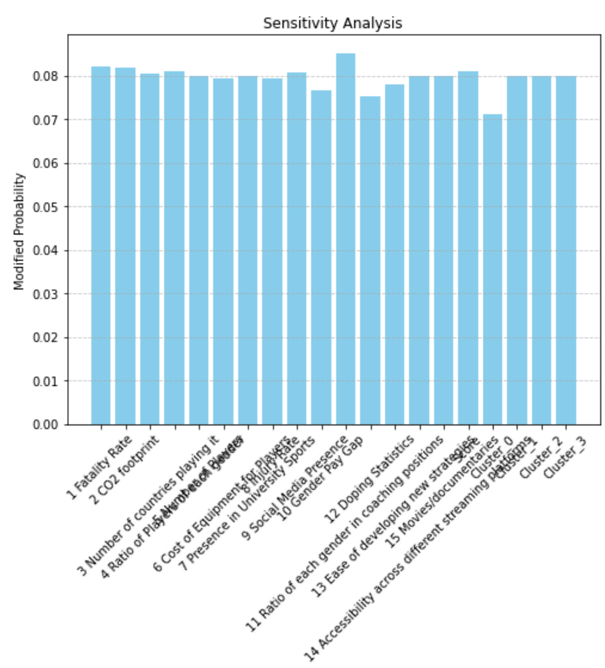
$$\Delta IF = 100p \cdot w\Delta v$$

Where:

1. ΔIF is the change in the impact factor value.
2. Δv is the change in the attribute value.
3. w is the attribute weight of the given attribute.

To evaluate the model's sensitivity, we analyzed a sample of 16 sports by systematically modifying each feature and measuring its impact on predictions. Using a loop, we altered each feature individually (e.g., increasing or decreasing its value by 10%) and calculated the resulting changes in probabilities. The results were visualized using a bar chart, and features were categorized based on their influence on the model's decisions. The following table summarizes the sensitivity analysis results. Following table shows the sensitivity analysis results.

Feature	Modified Probability
11 Ratio of each gender in coaching positions	0.085123572
1 Fatality Rate	0.082094267
2 CO2 footprint	0.081811408
4 Ratio of Players of each gender	0.081094429
Score	0.080955033
9 Social Media Presence	0.08063535
3 Number of countries playing it	0.080325977
5 Number of Players	0.080003961
14 Accessibility across different streaming platforms	0.080003098
15 Movies/documentaries	0.080002908
7 Presence in University Sports	0.080002545
Cluster_1	0.079997787
Cluster_2	0.079997787
Cluster_3	0.079997787
8 Injury Rate	0.079401755
6 Cost of Equipment for Players	0.079398691
13 Ease of developing new strategies	0.078076327
10 Gender Pay Gap	0.076751008
12 Doping Statistics	0.07538223
Cluster_0	0.071154472



The sensitivity analysis revealed that the **Ratio of each gender in coaching positions** (0.085124) was the most influential feature, significantly impacting the model's predictions, with even minor changes causing notable probability fluctuations. Other highly sensitive features included the **Fatality Rate** (0.082094) and the **Ratio of Players of each gender** (0.081094), both playing a key role in shaping the model's decisions. Moderately influential features, such as **CO2 Footprint** (0.081811), **Social Media Presence** (0.080635), and **Movies/documentaries** (0.080003), showed some impact on the predictions but were less critical than the top features. In contrast, features like **Doping Statistics** (0.075382) and **Gender Pay Gap** (0.076751) exhibited low sensitivity, indicating that changes in these variables had minimal effect on the model's outcomes.

Model Robustness

We added noise to the input features in order to check the robustness of our model; in addition, we obtained our model's prediction for the noisy data and measured metrics like accuracy. Repeated for increasing noise, represented as a normal distribution with standard deviation σ, from 0.01 to 0.5, the model shows consistent results in its performance, and the accuracy ranges from 0.75 to 0.875 on a small sample size of 16 sports. These results reflect that the model is quite robust under different noise conditions. Following table shows the model robustness results in detail.

Noise Level	Accuracy with Noise
0.01	0.875
0.05	0.75
0.1	0.875
0.2	0.8125
0.5	0.75

Our model is robust overall, maintaining high accuracy despite noise. It demonstrates resilience to minor perturbations and performs acceptably even under significant noise.

5.2 Strengths and Weaknesses

The strengths of our model are embedded in the integration of subjective and objective methodologies, which guarantee the balancing of the assessment process. By integrating AHP with the Entropy Method, the model reduces subjectivity and empowers data-driven insight to improve robustness. The ordering of variables in a hierarchical structure can make decisions systematic and easier to evaluate complex factors like gender equity, environmental impact, and accessibility. Further, the model's flexibility accommodates a wide range of variable types, allowing adaptation to continually changing and diverse criteria. Use of the Entropy Method weights quantitatively according to actual variability of data, while machine learning linearly enhances predictive accuracy with combined scores and clusters. Again, K-means clustering adds further insight by grouping sports similar in nature with one another, enabling prioritization and in-depth analysis. Such scalability and transparency ensure relevance for future evaluations, offering a reproducible framework that changes with new data and priorities.

However, the model has its weaknesses. Reliance on AHP introduces subjectivity through pairwise comparison, probably leading to bias from evaluators' experience and judgment. The model is also very sensitive to availability and quality issues of quantitative data, which may be significantly different for the variables of social media presence and gender pay gaps. Implementation challenges can also inhibit the integration of AHP, Entropy, clustering, and machine learning unless sufficient expertise is available. Small changes in weights or the scoring process can often lead to significant differences in the output, making the model sensitive. While the decision matrix and clustering are most valuable, interpretation for a non-technical decision-maker could require additional effort or visualization. The model also assumes linear relationships between variables and outcomes, which may over-simplify the nuanced interplay of factors. While the model works great with quantitative data, some qualitative factors, such as cultural significance or subjective appeal, might be hard to include, and there is a risk of overfitting the machine learning classifier if the training data is not representative.

Considering these issues, several improvements can be made. Sensitivity analyses would have to be done in order to understand how changes in weight or changes in data affect outcomes and improve robustness. Improved visualizations and summarizing outputs could make the results of the above more accessible to a decision-maker. Updates in the dataset would require periodical retraining of the machine learning classifier to keep the model relevant. Such enhancements would further reinforce the effectiveness and versatility of the model; it will be even more reliable as a guiding tool in decision-making on Olympic sports, disciplines, and events.

6 Letter to the IOC

Dear IOC,

We are writing today to recommend a novel solution to the tedious and difficult process of selecting which sport to include in the Olympics. Oftentimes, the inclusion and exclusion of certain sports can bring great backlash, and we know it is your goal to make the Olympics as successful as possible. Therefore, recognizing the various factors that go into the decision of whether a sport is included in a certain Olympics, we have created a mathematical model to assign scores to sports, allowing you to more easily make a decision. Our model comprehensively covers the various important factors, and is aligned with the values of the IOC, which are Popularity and Accessibility, Gender Equity, Sustainability, Inclusivity, Relevance and Innovation, and Safety and Fair Play.

In our model, we select quantifiable properties of each factor to represent in our model. For example, we used fatality rate and injury rate as our statistics for “Safety”. In total, we found 15 factors to quantify the 5 core values of the IOC selection criteria. We then applied a model called the AHP model, which helps to make decisions in the face of multiple factors. The model works in two steps. In the first step, factors are compared against each other for each sport, indicating their importance. For example, in a sport such as boxing, the fatality rate would be weighted much higher than carbon emissions, because the fatality rate is a large concern in such a risky sport. On the other hand, in a sport such as tennis, fatality rate would not be weighted as much, because death is not a major concern. Once weights for each of the 15 factors have been generated in the first step, the second step involves using the real life statistics for that sport to give it a score according to the weighted factors. For example, in boxing, we would look at the fatality rate per 1000 athlete exposures, and multiply that by the weighting on the fatality rate factor for boxing. That would give us the score for boxing in the fatality rate category. We would then subtract that score from the overall score from boxing, since fatality rate is a value that we want to minimize. We then do this for each of the 15 factors, adding factors that are desirable, and subtracting factors that are not, to arrive at the final score for boxing. We then repeat this with each sport to assign a score for each sport.

To decide which sports to include or exclude from the Olympics, you can just look at the score of the sport. For example, you could look at the top scoring sports currently not in the Olympics to decide if any should be added, or look at the lowest scoring sports currently in the Olympics to decide which should be removed. We hope that this model will be helpful in the decision making process.

Sincerely,
HiMCM Team 15276

7 References

[1] Factsheet: The programme of the Games of the Olympiad, ©International Olympic Committee, June 2024. Accessed online: https://stillmed.olympics.com/media/Documents/Olympic-Games/Factsheets/The-programme-of-the-Games-of-the-Olympiad.pdf

[2] David O. Manz, Thomas W. Edgar. "Logistic Regression." Logistic Regression - an Overview | ScienceDirect Topics, 2017, https://www.sciencedirect.com/topics/computer-science/logistic-regression

[3] Mark A. Lewis, Yanyu Xiao, Russell Greiner. "Evaluation of Machine Learning Methods for Predicting ... - Ualberta.Ca." Evaluation of Machine Learning Methods for Predicting Eradication of Aquatic Invasive Species, 2018, https://www.math.ualberta.ca/~mlewis/Publications%202018/Xiao2018_Article_EvaluationOfMachineLearningMet(1).pdf

[4] International Journal of Computer Science and Information Security, IJCSIS, Vol. 7, No. 1, pp. 292-295, January 2010, USA